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10. Exploring the Intersection of Large Language Models (LLMs) and Explainable AI (XAI): A Systematic Literature Review

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Abstract

*In the evolving domain of natural language processing (NLP), the emergence of Large Language Models (LLMs) like Generative Pre-trained Transformer (GPT), Bidirectional Encoder Representations from Transformers (BERT), and their advanced successors has markedly transformed text generation and comprehension. These models, leveraging deep learning and transformer architectures, have significantly enhanced NLP capabilities. Concurrently, the importance of Explainable AI (XAI) in elucidating the complexities of AI models, particularly LLMs, has gained prominence, especially for decision-making processes. This research-in-progress paper, structured under the PRISMA guidelines, provides initial insights into the confluence of LLMs and XAI. It explores two critical inquiries: (1) Which XAI methods are utilized to understand LLMs? and (2) **How are LLMs leveraged for their explanatory capabilities?** Publications from the four essential databases—Web of Science, Scopus, Arxiv, and IEEE Explore—are processed for this systematic review. Initially comprising 96 articles, the selection was refined to 21, focusing on studies that integrate LLMs and XAI with empirical findings. This review highlights current XAI research primarily relies on agnostic, local, quantitative methods like SHAP and LIME, often combined, to comprehend AI decision-making. **LLMs like ChatGPT excel in providing explanation of XAI outputs, revealing a gap in direct LLM utilization as explainer methods.***

Keywords: Large Language Models, Explainable AI, ChatGPT, SHAP, Systematic Literature Review

1. Introduction

The introduction of the Transformer architecture has significantly advanced Large Language Models (LLMs) such as the Generative Pre-trained Transformer (GPT) and Bidirectional Encoder Representations from Transformers (BERT), revolutionizing machine language processing (Harrer et al., 2023). These LLMs, grounded in deep learning, often function as ‘black boxes’ obscuring the processes behind their decision-making. This opacity is a critical concern as LLMs increasingly influence important decisions, where transparency and reliability are paramount (Islam et al., 2022). In contrast to LLMs, simpler models like regressions and decision trees offer greater interpretability but lack the capability to process complex, unstructured data effectively (Malhotra & Jindal, 2023). This dichotomy between performance and interpretability has underscored the importance of Explainable Artificial Intelligence (XAI). Since 2017, XAI has aimed to demystify the workings of complex ML

models, including LLMs (Islam et al., 2022). The urgency for XAI is heightened by LLMs' potential to produce erroneous or 'hallucinatory' outputs, which can inadvertently spread misinformation (Ji et al., 2023). Thus, XAI emerges as an essential tool in ensuring that the deployment of LLMs is both responsible and trustworthy.

However, transparency, a key aspect of responsible AI, remains underexplored in discussions on LLMs (Liao & Vaughan, 2023). To the best of our knowledge, this is the first systematic literature review addresses this pivotal aspect of AI by exploring LLMs in the XAI context, examining their roles as both subjects of other XAI methods and model explainers.

2. Methodology

This systematic review adheres to the 2020 PRISMA guidelines (Page et al., 2021). The main focuses of the research-in-progress study are as follows: the development of XAI methodologies aimed at elucidating the complex internal mechanisms of LLMs; and the **utilization of LLMs as instrumental explanation tools**.

We **employ a thorough literature search strategy** across four key academic databases: Web of Science (WoS), Scopus, Arxiv, and IEEE Xplore Digital Library. The search, focused on XAI within the context of LLMs, utilized carefully selected keywords. WoS (Birkle et al., 2020) and Scopus (Boyle & Sherman, 2006) were chosen for their extensive bibliographic collections and academic influence, while IEEE Xplore's vast repository offering over 6 million documents in technology and engineering and Arxiv's role as a leading open-access platform (Ginsparg, 2021) were pivotal for capturing the latest and most innovative researches enabled via the arXiv python API. This approach ensured a comprehensive and up-to-date review, making it indispensable for an in-depth exploration of LLMs in these fields.

Each keyword within a group is combined using the OR operator, and different groups are linked with the AND operator to create an extensive search string. The final line of Table 1 shows how keywords from various groups are amalgamated to construct a search query deployed across all four bibliographic databases, a search of titles, abstracts, and indexing terms for all available years since all relevant papers in this field are of recent publication.

The keywords for query formulation is displayed below (see Figure 1):

Group 1: Explainable AI Related Keywords	Group 2: Large Language Model Related Keywords
XAI OR explainable AI OR explainable artificial intelligence OR black-box OR post-hoc explainability OR intrinsic explainability OR counterfactual approximation OR counterfactual explanation OR feature importance	LLM OR chatgpt OR GPT-* OR generative AI OR large language model* OR pretrained language model*
Search Query: (GROUP 1) AND (GROUP 2)	

Table 1: Selected keywords in different groups.

We established **criteria for both inclusion and exclusion to ensure a focused and relevant review**. This rigorous approach allowed us to refine the selection process, securing a collection of articles that precisely aligned with our research aims.

Details of inclusion and exclusion criteria are listed in Table 2:

Inclusion criteria
1) The paper claims that an LLM and XAI method used.
2) The paper is published in either a journal or conference proceedings in selected databases except arxiv.
3) The paper is in English.
4) The paper with accessible full text.
Exclusion criteria
1) The paper mentioning the used of LLMs without describing an XAI method.
2) The paper applying LLMs for purposes non-related with XAI.
3) The paper released in a workshop.
4) The paper without a clear demonstration of results for LLMs.

Table 2: Inclusion criteria and Exclusion criteria.

The PRISMA search methodology is displayed in Figure 1:

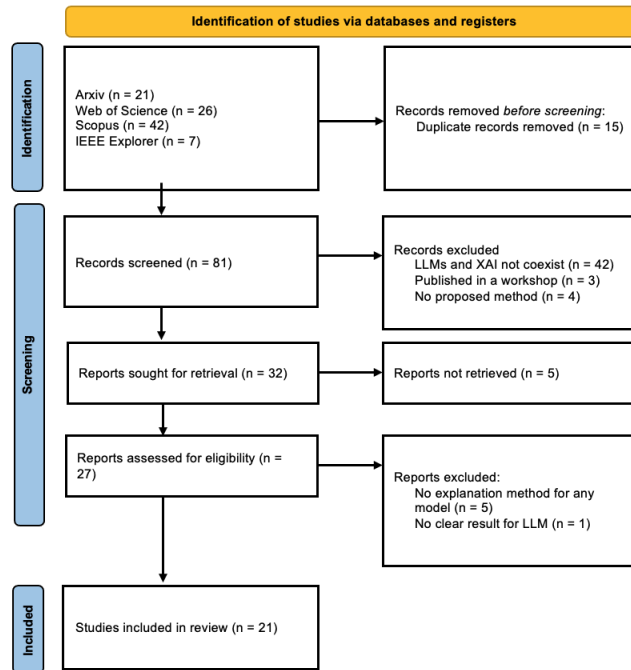


Figure 1: PRISMA search methodology.

(Adapted from Page et al., 2021)

The search across these databases yielded a total of 96 studies based on the selected keywords. Duplicate entries from different databases were meticulously examined, retaining only unique records for each primary sample. During this duplicate removal process, 15 studies were excluded. Through a manual selection process focusing on titles and abstracts, the initial set of 81 studies was narrowed down to 32 unique and pertinent articles. Then, the count was further reduced to 27 papers, since access to 5 articles could not be retrieved. Ultimately, after excluding 6 articles based on the defined exclusion criteria through full-text review, 21 articles remained for consideration.

3. Results

We initially would like to introduce a structured summary of recent scholarly contributions at the intersection of LLMs and XAI to identify which XAI methods utilized to explain the inner working of LLMs and what roles LLMs taken in this studies.

The summary of recent studies for the SLR is displayed in Table 3:

Author & Year	XAI Method	Utilized LLM	Role of LLM
Ali & Kostakos, 2023	Local Interpretable Model-agnostic Explanations (LIME), Shapley Additive Explanations (SHAP)	ChatGPT (GPT-3.5)	XAI Output Translator
Bhattacharjee et al., 2023	LLM, Counterfactual Explanation (CFE)	text-davinci-003, ChatGPT, GPT-4, LLaMA2 7B, LLaMA2 13B - DistilBERT	Explanation Generation, LLM as Subject Model
Cha & Lee, 2023	SHAP	BART	LLM as Subject Model
Chang et al., 2023	LIME	GPT-2, MASS, BART	LLM as Subject Model
Chen et al., 2023	Knowledge Graph (KG), Graph Attention Network (GAT) for why-choose, why-not choose explanations, and reason elements	ROBERTA - GPT 3.5, GPT-4	LLM as Subject Model, XAI Output Translator
Demertzis et al., 2023	SHAP, CFE	ChatGPT	XAI Output Translator
Feldhus et al., 2023	Saliency Map	GPT-3.5	XAI Output Translator
Feleki et al., 2023	Heatmap	GPT-3.5	XAI Output Translator
Gat et al., 2023	LLM, CFE	ChatGPT, T5 - DistilBERT, BERT, ROBERTA, Llama2-7b, Llama2-13b	Explanation Generation, LLM as Subject Model
Kroeger et al., 2023	LLM, LIME, SHAP	ChatGPT (GPT-3.5, GPT-4)	Explanation Generation
Lucchese et al., 2023	Embedding Analysis for distance between query and output token embeddings	BERT, ROBERTA, ELECTRA	LLM as Subject Model
Malhotra & Jindal, 2023	LIME, SHAP	BERT, DistilBERT, RoBERTa, MentalBERT, PsychBERT, PHSBERT	LLM as Subject Model
Martens et al., 2023	SHAP, CFE	ChatGPT (GPT-4)	XAI Output Translator
Metzger et al., 2023	Explainable Reinforcement Learning for Decomposed Interestingness Elements (XRL-DINE)	ChatGPT (GPT-3.5)	XAI Output Translator
Mitrović et al., 2023	SHAP	DistilBERT, GPT-2	LLM as Subject Model
Nazary et al., 2023	Feature Importance	ChatGPT (GPT-3.5)	LLM as Subject Model
Sarti et al., 2023	Feature Importance (Perturbation-based, Internal-based, Gradient-based)	GPT-2	LLM as Subject Model
Susnjak, 2023	SHAP, CFE	ChatGPT	XAI Output Translator
Tursun et al., 2023	Heatmap	ChatGPT	XAI Output Translator
Wellawatte & Schwaller, 2023	LIME, SHAP	ChatGPT (GPT-4)	XAI Output Translator
Yang et al., 2023	Human Knowledge	GPT-3	LLM as Subject Model

Table 3: Exploring the Intersection of LLMs and XAI: A Summary of Recent Research.

3.1 Understanding the Inner Working of the LLMs

In the landscape of AI research, two primary explanatory approaches are recognized: local, focusing on individual model decisions, and global, which elucidates a model's overall behavior. SHAP is particularly noteworthy for its ability to offer both local and global insights (Wellawatte & Schwaller, 2023). Its game theory foundation ensures an equitable attribution of contribution across features, making SHAP a highly effective tool for delving into the nuanced dynamics of AI models (Mitrović et al., 2023). In contrast, LIME and CFE are tailored towards local interpretability, offering clarity on specific predictions and hypothetical decision alterations, respectively. Despite SHAP's efficacy as a standalone tool—employed independently in 45% of studies—the trend towards integrating it with methods like LIME and CFE is emerging in about 40% of research. This integration amplifies the interpretability scope by combining SHAP's broad analytical reach with the detailed, scenario-specific insights provided by LIME and CFE. Such a composite approach not only enhances the granularity of local explanations but also enriches the understanding of global model dynamics.

Most of the employed methods are local explainers which focus on providing explanations for individual predictions. This trend towards local explanations may stem from the growing need to understand specific decision-making instances in AI, where individual decisions can have significant consequences (Martens et al., 2023). Moreover, methods like SHAP, LIME, CFE, and KG, which tend to offer more quantitative insights, are prevalent. This preference could be due to the increasing demand for measurable and data-driven explanations in AI, especially in research and development environments where quantifiable insights are crucial for model evaluation and improvement. Additionally, model-agnostic methods like LIME and

SHAP are broadly applicable across various models, favored in over 40% of studies for their flexibility and universal insights. Conversely, model-specific techniques, such as KG, GAT, heatmaps, offer precise, tailored insights into the specific workings of particular model architectures (Tursun et al., 2023). This highlights AI research's dual needs: for versatile tools that broadly interpret AI behaviors and for specialized methods that deeply analyze specific models.

Moreover, the analysis of 21 articles on XAI methods shows that while around approximately 48% did not specify their target group, likely indicating a more general audience, the specified audiences predominantly cater to non-technical groups such as judicial officers, students, chemist, and healthcare practitioners, with a smaller portion targeting technical professionals like programmers and data scientists.

3.2 LLMs as Explainers

In the XAI realm, LLMs such as ChatGPT play a pivotal role in translating complex XAI method outputs into digestible narratives, bridging the gap between intricate analytical results and their practical implications. This translation is key for stakeholders without technical expertise, fostering transparency and trust in AI systems (Demertzis et al., 2023; Martens et al., 2023). These models proficiently convert intricate data from approaches like heatmaps or Shapley value distributions into lucid explanations embedded with contextual understanding, thereby making the AI decision-making process more intelligible.

Furthermore, ChatGPT extends beyond translation to actively create explanations. It generates counterfactual narratives and isolates the most influential features affecting the model's decisions, offering insight into the model's rationale (Bhattacharjee et al., 2023). Moreover, LLMs themselves are now under XAI investigation, with BERT and GPT variants undergoing scrutiny. Applying XAI techniques to these models is essential for maintaining transparency and trust, particularly in sensitive sectors like healthcare (Yang et al., 2023; Malhotra & Jindal, 2023). While LLMs are predominantly engaged in translating XAI outputs and are increasingly the focus of XAI studies, the direct application of XAI tools to analyze LLMs signals a shift towards a deeper comprehension of AI internals, as crucial as the external applications. This trend underscores the dual utility of LLMs: as pragmatic tools for explanation generation and as subjects for in-depth analytical examination, promoting the development of transparent and interpretable AI systems. Despite their efficacy in interpreting outputs from various XAI techniques, the exploration of LLMs as independent explainers has not been fully realized, indicating a prospective area for growth within XAI research (Kroeger et al., 2023; Bhattacharjee et al., 2023).

4. Conclusion and Future Work

Based on this research-in-progress, it is evident that current research in XAI predominantly relies on agnostic, local, quantitative methods like SHAP and LIME, often in combination, to delve into the intricacies of AI decision-making processes. Moreover, LLMs, such as ChatGPT play a pivotal role in providing XAI output translator, making complex AI outputs accessible. However, a noticeable gap remains in the direct utilization of LLMs as explainer methods in explaining other predictive models. To further enhance the development of transparent AI systems, future work will expand its scope using the forward-backward method and additional dimensions such as application areas, XAI evaluation metrics, and prompting techniques.

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